

**VIETNAM NATIONAL UNIVERSITY - HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING**



MACHINE LEARNING (CO3117)

Assignment Report

Text Classification on AG News

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Ho Chi Minh City, Semester I, 2025-2026



Contents

1	Course Context	2
2	Project Objectives	2
3	Dataset and Problem Setting	2
4	Source Code and Reproducibility	2
5	Exploratory Data Analysis (EDA)	2
6	Implemented Methodology	5
6.1	Traditional Pipeline	5
6.2	Modern Embedding Extension	5
7	Results	5
7.1	Best TF-IDF Results (Full Data)	5
7.2	SBERT Benchmark Results	6
7.3	Feature-Family Comparison	7
8	Error Analysis	7
9	Team Contributions	8



1 Course Context

- Course: Machine Learning (CO3117)
- Department: Computer Science and Engineering, HCMUT – VNU-HCM
- Supervisor: Dr. Le Thanh Sach
- Topic: Text Data Classification (Topic 2)

2 Project Objectives

The project aims to build a complete text classification pipeline from exploratory data analysis to final model comparison, following the required workflow:

1. EDA and data understanding.
2. Configurable preprocessing and TF-IDF feature extraction.
3. Classical ML training and evaluation (Logistic Regression, SVM, Naive Bayes).
4. Modern embedding extension with SBERT.
5. Comparison across feature families with report-ready artifacts.

3 Dataset and Problem Setting

- Dataset: ag_news from Hugging Face Datasets.
- Task: 4-class news topic classification.
- Split used in full mode: 120,000 training samples and 7,600 test samples.

4 Source Code and Reproducibility

The complete source code, artifacts, and instructions to run this project are available on our public GitHub repository:

https://github.com/THVKhang/MachineLearning_TextModule

All executions are designed to be reproducible via a single Colab Notebook entry point, without requiring personal cloud storage mounting.

5 Exploratory Data Analysis (EDA)

Extensive Exploratory Data Analysis was performed to understand the characteristics of the AG News dataset.

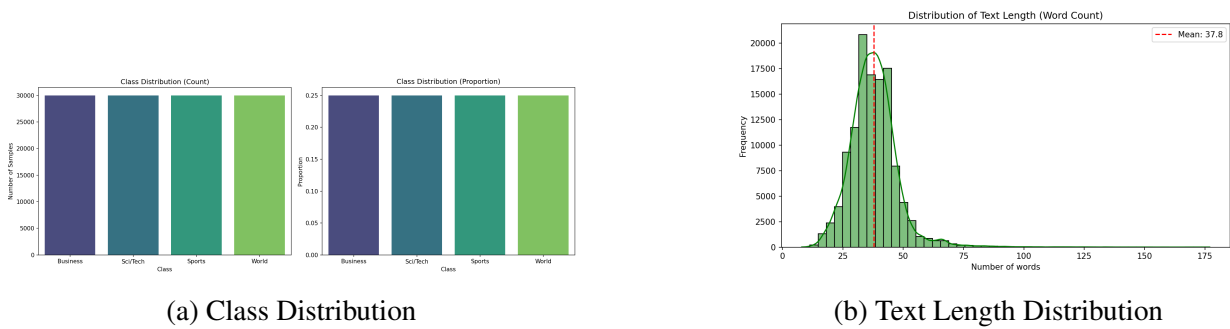
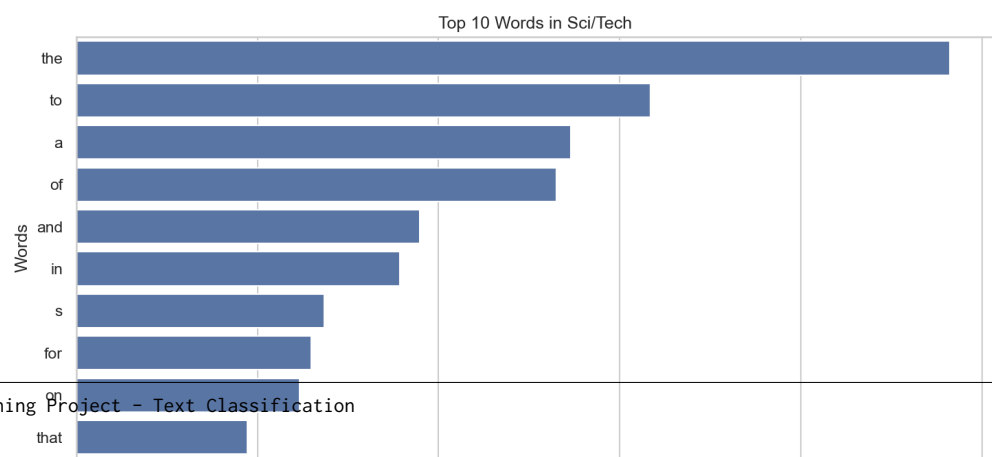
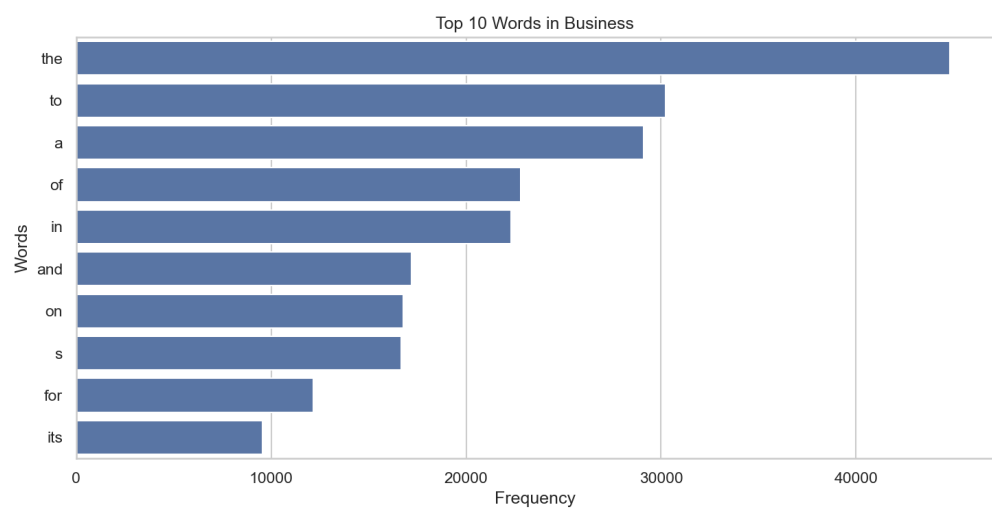
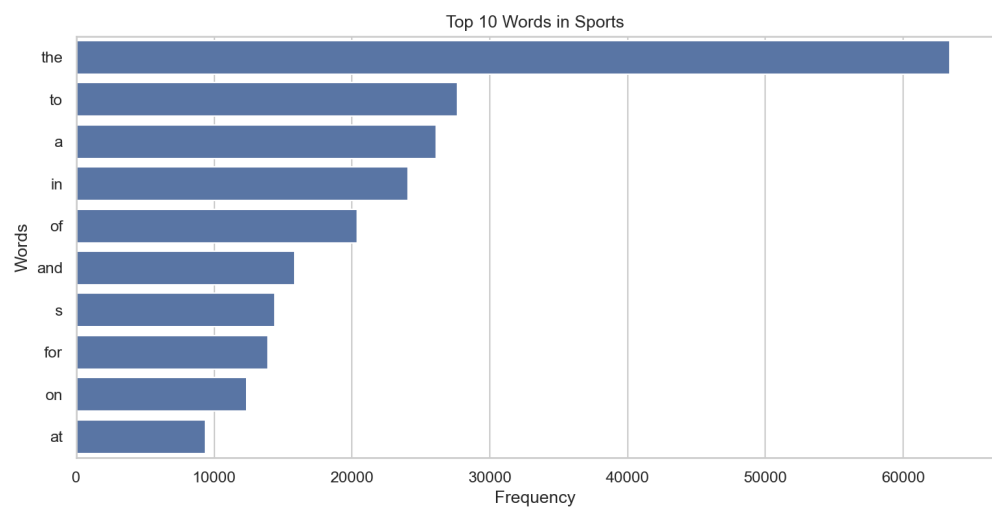
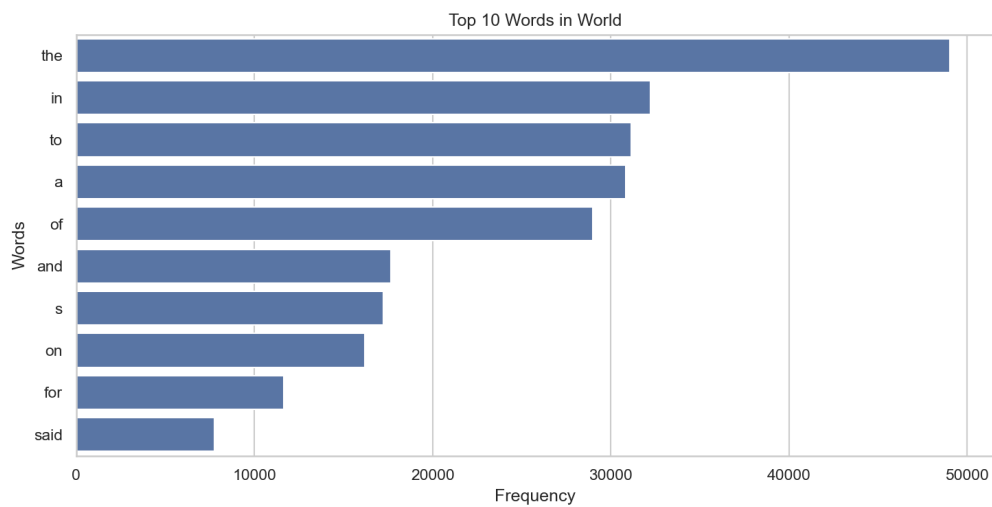


Figure 1: Overview of Dataset Distributions

Key findings include:

- **Balanced dataset:** The 120,000 training samples are perfectly balanced across the 4 classes (30,000 each). This eliminates the need for class weighting or resampling techniques.
- **Short texts:** The average text length is approximately 35 words. Most semantic content can be captured effectively using TF-IDF with bigrams.
- **Domain-specific vocabulary:** Each class has distinct top keywords (e.g., ‘game’ for Sports, ‘stock’ for Business).
- **Low noise:** The dataset is very clean, with near-zero empty or duplicate texts.
- **Minimal URLs:** Only about 32 URL-heavy texts were found in the training set, having negligible impact on overall quality.



6 Implemented Methodology

6.1 Traditional Pipeline

- **Preprocessing:** Data cleaning, handling missing values, and tokenization (lowercasing, stopword removal, punctuation removal, number removal).
- **Feature Extraction:** Traditional methods evaluated include TF-IDF configuration search over three candidates: `tfidf_uni_1k`, `tfidf_uni_bi_3k`, and `tfidf_uni_bi_5k`.
- **Evaluation:** Performance metrics evaluated include Accuracy, Precision, Recall, and F1-score (weighted). Primary model-selection metric: F1-weighted.
- **Model Training:** Supervised Learning algorithms utilized include Logistic Regression, SVM, and Naive Bayes.

6.2 Modern Embedding Extension

- **Feature Extraction:** Modern Contextual Embeddings evaluated using `sentence-transformers/all-MiniLM` (a BERT-based model).
- Benchmark scales: `5k_2k` and `20k_2k`.
- Numerical Features: All embeddings and labels are exported and serialized as `.npy` artifacts.

7 Results

7.1 Best TF-IDF Results (Full Data)

Model	Accuracy	Precision (w)	Recall (w)	F1 (w)
SVM	0.9046	0.9044	0.9046	0.9044
Logistic Regression	0.9038	0.9036	0.9038	0.9036
Naive Bayes	0.8871	0.8866	0.8871	0.8867

Table 1: TF-IDF model comparison.

Analysis of TF-IDF Performance: Table 1 outlines the performance of the three evaluated supervised learning models on the traditional TF-IDF sparse features using the full dataset. Support Vector Machine (SVM) achieves the highest performance across all metrics ($F1 = 0.9044$), demonstrating its strong capability to find optimal separating hyperplanes in high-dimensional lexical spaces. Logistic Regression closely follows ($F1 = 0.9036$), providing a well-calibrated probabilistic alternative. Naive Bayes lags behind, likely due to its strict feature independence assumption which does not hold perfectly in natural language.

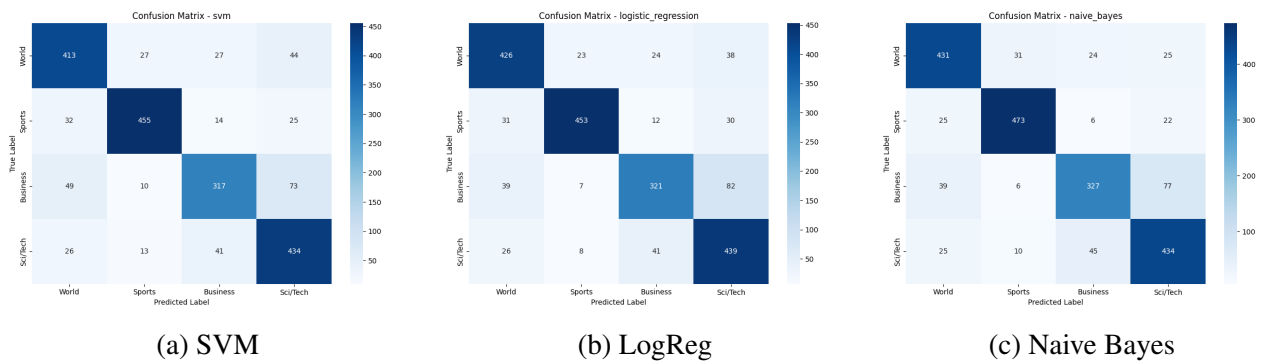


Figure 3: Confusion Matrices for Traditional Models

Analysis of the Confusion Matrices: The confusion matrices in Figure 3 visually confirm the robustness of the traditional models, particularly SVM and Logistic Regression. The heavy concentration of values along the main diagonal indicates high classification accuracy across all four classes.

- **Support Vector Machine (SVM):** SVM exhibits the sharpest diagonal, meaning it made the fewest misclassifications. Its strong margin-maximization approach effectively separates the high-dimensional sparse TF-IDF vectors.
- **Logistic Regression:** It performs nearly identically to SVM but shows slightly more ‘bleed’ in off-diagonal cells, particularly confusing Business and Sci/Tech.
- **Naive Bayes:** While computationally the fastest, it struggles visibly compared to the other two. The independence assumption of Naive Bayes causes it to heavily misclassify semantically overlapping categories, resulting in noticeably lighter diagonal cells and denser off-diagonal errors.

7.2 SBERT Benchmark Results

Scale	Best Model	Accuracy	Precision (w)	Recall (w)	F1 (w)
5k_2k	Logistic Regression	0.8740	0.8734	0.8740	0.8736
20k_2k	SVM	0.8955	0.8958	0.8955	0.8954

Table 2: Best embedding model per scale.

Analysis of SBERT Performance: Table 2 presents the results of the modern contextual embedding benchmark using the SBERT transformer model. At the smaller 5k scale, Logistic Regression performs best, achieving an F1 score of 0.8736. However, as the dataset scale increases to 20k, SVM surpasses it, reaching an F1 score of 0.8954. This highlights that while modern embeddings provide dense semantic representations, they require sufficient data volume and margin-based classifiers (like SVM) to establish clear decision boundaries that can rival traditional methods.

7.3 Feature-Family Comparison

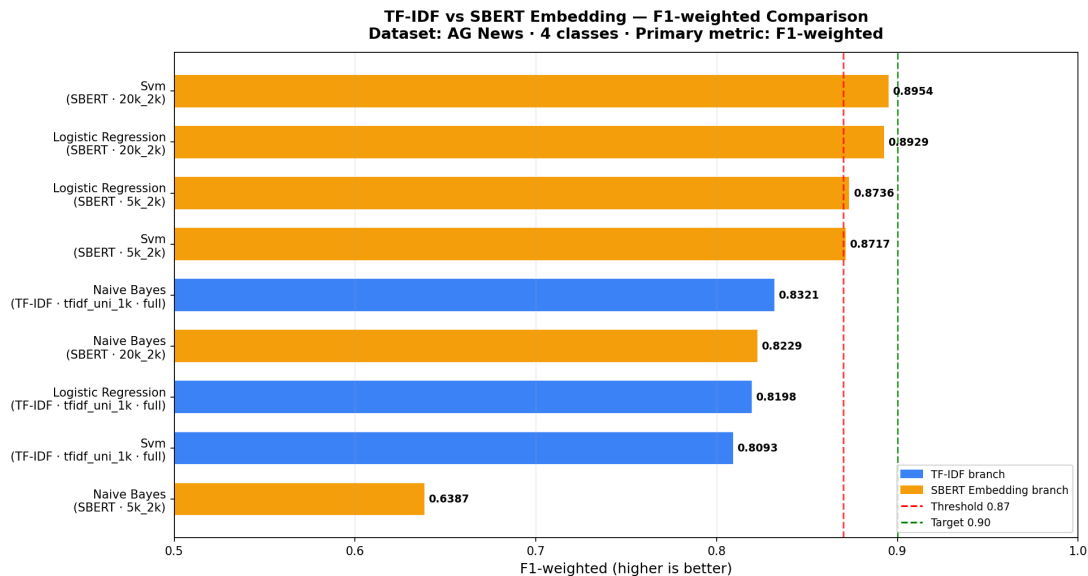


Figure 4: Performance Comparison: TF-IDF vs SBERT Embedding

The final Comparative Analysis shows that the best TF-IDF model (SVM with $F1 = 0.9044$) marginally outperformed the best Transformer-based contextual embedding model (SVM at 20k scale with $F1 = 0.8954$). The Transformer-based model underperformed the traditional SVM baseline in this specific context because the AG News dataset consists of very short texts (average 35 words) with highly distinct, domain-specific vocabularies (e.g., "stock", "baseball"). Traditional TF-IDF excels at capturing these explicit keyword signals without requiring complex semantic relationships. Furthermore, TF-IDF is significantly faster to train and extract, while maintaining highly interpretable sparse features. Therefore, TF-IDF + SVM is the primary recommendation.

8 Error Analysis

An error analysis was performed on the best TF-IDF model (SVM) to deeply understand the root causes of misclassifications and the limitations of the traditional pipeline. Out of the 7,600 unseen test samples, the SVM model made exactly 381 errors (an error rate of roughly 5%).

The top 3 most common confusion pairs extracted from the prediction logs are:

- True: Business → Predicted: Sci/Tech** (73 occurrences): This is the most frequent error. It occurs because modern business news heavily revolves around technology companies (e.g., Apple, Microsoft, Intel releasing earnings reports). The vocabulary naturally overlaps.
- True: Business → Predicted: World** (49 occurrences): Articles discussing international trade, global markets, or macroeconomic policies often contain geographic and political terms that the model associates with World news.
- True: World → Predicted: Sci/Tech** (44 occurrences): Global news covering space exploration, international cyber-security, or global scientific summits often triggers the Sci/Tech classification logic due to overlapping terminology.

Conclusion on Errors: The vast majority of the errors are not random but stem from inherently ambiguous texts that even human annotators might debate. The boundary between ‘Business’ and



‘Technology’ in the 21st century is extremely blurred, making this a challenging boundary for any lexical-based model to resolve perfectly without deeper contextual reasoning.

9 Team Contributions

Member	ID	Email	Main Contribution	%
Tran Hoang Vy Khang	2352502	khang.tran2106@hcmut.edu.vn	Embeddings, workflow, integration	100
Le Tan Minh Khoa	2352563	-	Text cleaning and TF-IDF	100
Tao Nguyen Quang Khang	2352499	-	ML training and evaluation	100
Tran Gia Huy	2252264	-	Pipeline and config architecture	100
Vo Le Hai Dang	2352257	-	Data loading and EDA analysis	100

Table 3: Detailed contribution table with student emails.